

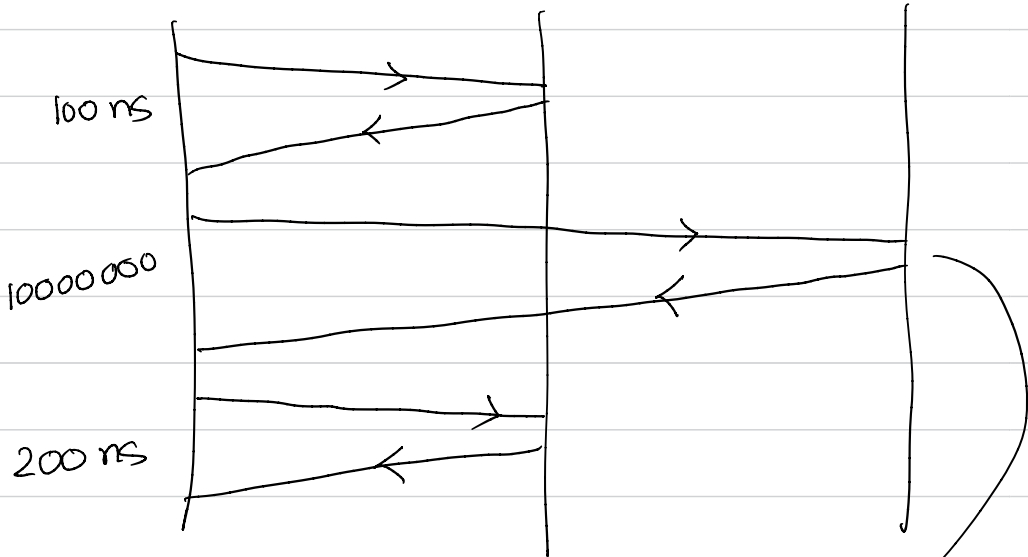
MID Solution /

1) a)

DMA controller

MP

I/O device



$$10000300 - (100 + 200) \downarrow$$
$$= 10000000 \text{ ns}$$

$$\text{data transfer rate} = \frac{1536}{1000000 \times 10^{-9}}$$
$$= 153600 \text{ bytes/sec}$$
$$= \frac{153600}{1024} = 150 \text{ KB/sec}$$

b)

$$\text{Burst mode} \rightarrow \frac{1536}{3} = 512 \text{ bytes}$$

$$\text{Cycle stealing mode} \rightarrow \frac{1536 \times 2}{3} = 1024 \text{ bytes}$$

$\therefore$ Burst mode $\rightarrow$

$$100 \text{ ns} + \frac{512 \times 10^9}{150 \times 10^{24}} \text{ ns} + 200 \text{ ns}$$

$$= 3333633.333 \text{ ns}$$

$\therefore$ Cycle stealing mode $\rightarrow$

$$\frac{1024}{4} \left(100 \text{ ns} + \frac{4 \times 10^9}{150 \times 10^{24}} \text{ ns} + 200 \text{ ns} \right)$$

$$= 6743466.667 \text{ ns}$$

$$\therefore \text{Total time} = 10077100 \text{ ns}$$

8 4 2 1

2 a)

$$\begin{array}{r}
 6BCD \rightarrow 0110 \quad 1011 \quad 1100 \quad 1101 \\
 5234 \rightarrow 0101 \quad 0010 \quad 0011 \quad 0100 \\
 \hline
 1011 \quad 1110 \quad 0000 \quad 0001
 \end{array}$$

$$ZF = 0$$

$$PF = 0$$

$$SF = 1$$

$$OF = 1$$

b) AX, BX, CX, DX

3) ^a Physical address

$$\begin{aligned}
 &= ((79E4H \times 10H) + 2A8CH) \\
 &= 7C8CCH
 \end{aligned}$$

$$11) 7C8CCH = (CS \times 10H) + 5A2DH$$

$$\Rightarrow CS = \frac{7CE9FH}{10H} \rightarrow \text{fraction}$$

CS can't be a fraction so invalid.

$$i) 7C8CC H = (CS \times 10H) + 5A3C$$

$$\Rightarrow CS = \frac{76E90 H}{10 H} = 76E9 H$$

$$ii) 7C8CC H = (2679H \times 10H) + IP$$

$$IP = 5613C H > FFFF H$$

But for $CS = 2679 H$ the address range is $\rightarrow (2679H \times 10H) + 0$
 $= 26790H$

$$\begin{aligned} &+ 0 \\ &(2679H \times 10H) + FFFF \\ &= 3678F H \end{aligned}$$

So $7C8CC H$ can't be reached
 $\therefore$ invalid.

$$\textcircled{b} \quad 2^{n_A} \times 32 = 128 \times 2^{20} \times 8$$

$$\Rightarrow 2^{n_A} = \frac{128 \times 2^{20} \times 8}{32}$$

$$\therefore n_A = \log_2 \left(\frac{128 \times 2^{20} \times 8}{32} \right)$$

$$= 25$$

$$\therefore A.B \rightarrow 25 \quad D.B \rightarrow 32$$

$$2^{26} \times m = 128 \times 2^{20} \times 8$$

$$\Rightarrow m = \frac{128 \times 2^{20} \times 8}{2^{26}}$$

$$= 16$$

$$\therefore A.B \rightarrow 26 \quad D.B \rightarrow 16$$

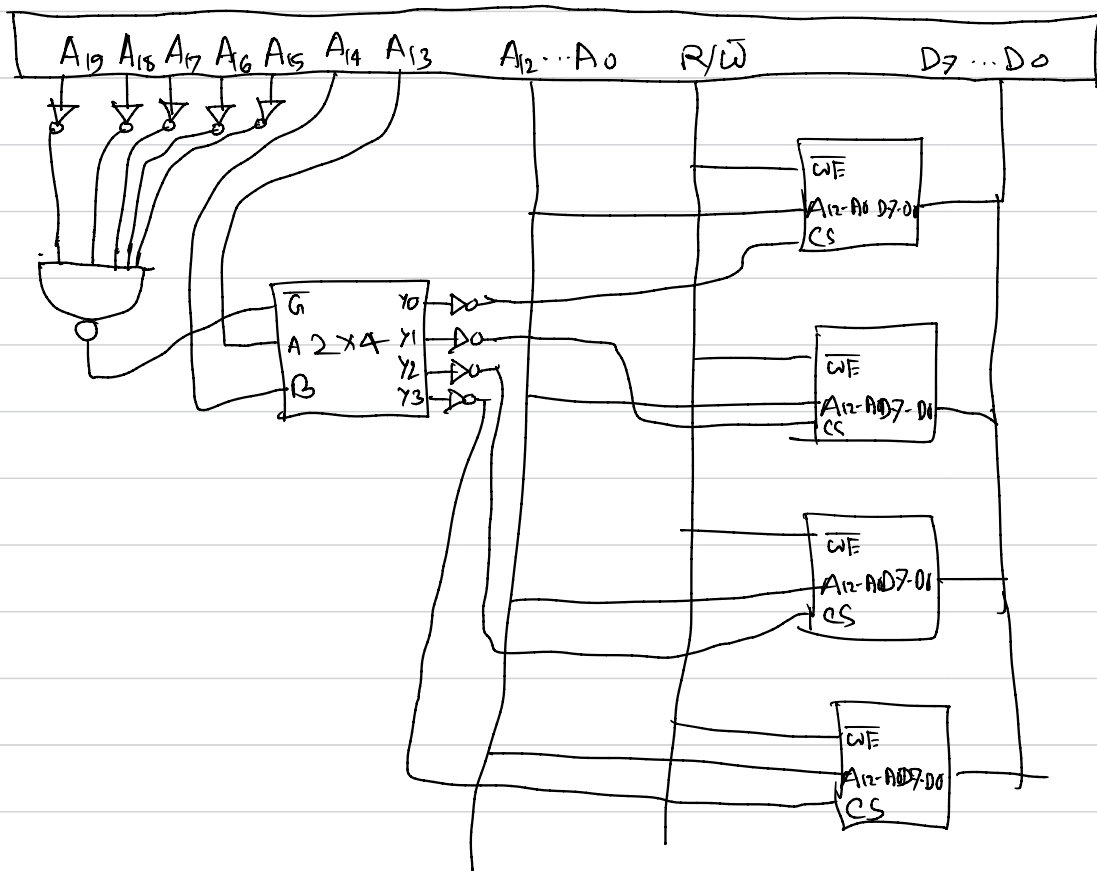
4/
①

MP $\rightarrow$ 20 bit A , 8 bit D

RAM $\rightarrow$ 13 bit A , 8 bit D

$$\text{RAM capacity} = \frac{2^{13} \times 1}{2^{10}} = 8 \text{ KB}$$

$$\therefore \text{RAM needed} = \frac{28}{8} \approx 4$$

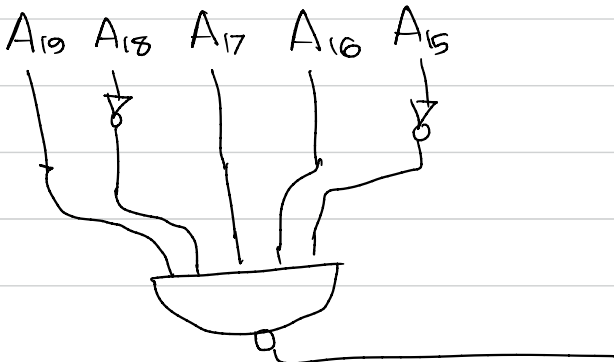


Address Range

$A_{19} \dots A_{15}$	$A_{14} A_{13} A_{12} \dots A_0$	
00...0	0 0 0...0	00000 H
00...0	0 0 11...1	01FFF H
00...0	0 1 00...0	02000 H
00...0	0 1 11...1	03FFF H
00...0	1 0 00...0	04000 H
00...0	1 0 11...1	05FFF H
00...0	1 1 00...0	06000 H
00...0	1 1 11...1	07FFF H
		01...1 06FFF H

b) $B0000H - B6FFFH$

$10110 \dots 0$ $10110110 \dots$



5/

a) EH x 4H — 38H

38H — IP (lower)

39H — IP (higher)

3AH — CS (lower)

3BH — CS (higher)